Unit Test Report

1. Purpose and scope

This report records the result of the project's automated Django test suite. The suite is run with python manage.py test against PostgreSQL and Redis.

2. Environment

Item	Value
Application	Online Bookstore (Team 2), Django 6.1.2, Django REST Framework 3.18.3 (requirements.txt)
Database and cache	PostgreSQL, Redis (as in docker-compose.yml)
Test runner	Django test runner (python manage.py test)
Coverage tool	coverage.py 7.17.2, branch measurement on
Run date	9 November 2026

3. Summary of results

Testing activity	Result
Full Django test suite	260 passed, 1 skipped (261 total)
End-to-end test	1 passed
Security tests	16 passed
Performance test	50 users, 30 seconds, 13,040 requests, 0 errors
Line coverage	92%
Branch coverage	85.45%

The one skipped test is the Celery worker test in notifications/test_worker.py, which is marked to run only in CI because it needs the Redis service there. All other tests passed. No test failed and none raised an error.

4. Test modules

The suite is organised in one or more test modules per app. Every module below belongs to the run summarised in section 3.

Test module	Area	Test classes
accounts.test_admin	Login, registration, lockout, roles, audit log, admin	AdminDashboardTests
accounts.test_audit	Login, registration, lockout, roles, audit log, admin	AdminAuditLoggingTests, AuditLogRetentionTests
accounts.test_rbac	Login, registration, lockout, roles, audit log, admin	RoleAssignmentTests, RolePermissionTests
accounts.test_rbac_endpoints	Login, registration, lockout, roles, audit log, admin	SellerManagementEndpointTests
accounts.test_registration	Login, registration, lockout, roles, audit log, admin	RegistrationTests
accounts.test_token_auth	Login, registration, lockout, roles, audit log, admin	TokenLoginTests, TokenAuthenticationTests, PasswordHashingTests
accounts.tests	Login, registration, lockout, roles, audit log, admin	AuthenticationModelTest, AccountLockoutTests
bookstore.test_e2e	End-to-end journey, error format, security	CustomerJourneyTests
bookstore.test_errors	End-to-end journey, error format, security	ErrorHandlerTests, ApiErrorFormatTests, LogRedactionTests
bookstore.test_security	End-to-end journey, error format, security	CustomerJourneyTests, FuzzTests, AccessControlTests, DataProtectionTests, OutputEscapingTests
cart.test_cart	Cart	CartTests
cart.tests	Cart	CartModuleTest
catalog.test_api	Catalog list, search, caching, detail, seller admin	BookListTests

Test module	Area	Test classes
catalog.test_cache_invalidation	Catalog list, search, caching, detail, seller admin	CacheInvalidationTests
catalog.test_caching	Catalog list, search, caching, detail, seller admin	CatalogCachingTests
catalog.test_detail	Catalog list, search, caching, detail, seller admin	BookDetailTests
catalog.test_search	Catalog list, search, caching, detail, seller admin	SearchFilterSortTests
catalog.test_seller_admin	Catalog list, search, caching, detail, seller admin	SellerAdminTests
catalog.tests	Catalog list, search, caching, detail, seller admin	BookModelTest
monitoring.tests	Monitoring	MonitoringTests
notifications.test_sms	Email, SMS, worker	SmsTests, OrderSmsTaskTests
notifications.test_worker	Email, SMS, worker	CeleryRedisWorkerTests
notifications.tests	Email, SMS, worker	NotificationTaskTests
orders.test_fulfilment	Checkout, payment, history, returns, fulfilment	TransitionRuleTests, FulfilmentTestCase, ChangeStatusTests, OrderAdminActionTests
orders.tests	Checkout, payment, history, returns, fulfilment	OrderModelTest, CheckoutAPITest, OrderHistoryAPITest, ReturnRefundRequestAPITest, RazorpayGatewayTest, PaymentVerificationIntegrationTests, RazorpayWebhookIntegrationTests
reviews.test_admin	Reviews API, moderation, admin	ReviewAdminTests
reviews.test_moderation	Reviews API, moderation, admin	ModerationTests
reviews.test_reviews	Reviews API, moderation, admin	ReviewApiTests
reviews.tests	Reviews API, moderation, admin	ReviewModelTest
storefront.test_book_detail	Storefront pages and address API	BookDetailPageTests
storefront.test_wishlist_page	Storefront pages and address API	WishlistPageTests
storefront.tests	Storefront pages and address API	StorefrontPageTests, AddressApiTests
wishlist.tests	Wishlist	WishlistTests

5. Code coverage achieved by the unit tests

Measure	Covered	Total	Percent
Statements	1475	1585	93.1%
Branches	229	268	85.45%
Combined (coverage.py total)	-	-	92.0%

Details per file are in the Code Coverage and Branch Coverage Report.

6. Defects found by testing

#	Defect found	Found by	Fix
1	The pull request failed in CI: PyJWT and bcrypt were missing from requirements.txt.	CI	Added both packages.
2	A very large min_price or max_price (for example 1e999) crashed the catalog with a server error.	Fuzz tests	Prices above the stored maximum are answered with a normal 400 error.
3	The verification link in the emailed message was cut at a line break, so the link did not work when copied.	Manual testing	A console email backend that prints the message without wrapping.
4	The 90th percentile was about 16 seconds under load on the development server.	Load test	Run under gunicorn with WhiteNoise (section 5).
5	A test file broke the test run after a merge conflict (a syntax error).	CI	The file was repaired and the run passed again.
6	The payment verification and webhook routes were lost when the invoice branch was merged, so main was red with 6 failing payment tests.	CI and local runs	The two routes were restored.

#	Defect found	Found by	Fix
7	Cached catalog and book pages could show old prices, stock and ratings after the data changed. Two tests failed locally.	Local test run	The cache is cleared whenever a book or review is saved or deleted, with new tests.

All defects above were fixed and re-tested. No known defect is open.

7. Conclusion

260 of 261 tests passed and 1 was skipped by design; there were no failures or errors. Line coverage is 92% and branch coverage is 85.45%.